Saheed Faremi

PhD researcher, deep generative models for EEG · Data scientist
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PhD researcher studying brain activity with deep learning, and a data scientist with five years of experience across software engineering and data science. Built AI systems that reduced manual document work by 85% and strengthened fraud detection across eight African markets.
Last updated November 2026

RESEARCH AREAS

Brain and behaviour: EEG microstates and brain-signal analysis; Neuroscientific knowledge discovery; Neuroimaging; Nonlinear time series analysis

Explainable and reproducible AI: Explainable Artificial Intelligence; Mechanistic Interpretability; Interpretable, reproducible machine learning

Applied AI and data systems: Deep learning for health and behaviour; Production ML, LLM and data systems

TECHNICAL SKILLS

Brain-signal analysis: PyTorch; scikit-learn; MNE; deep generative models; large-scale model evaluation

Production ML & LLMs: Azure OpenAI; retrieval-augmented generation (RAG); fraud and anomaly detection; transformer fine-tuning; document generation and automation

Software & cloud engineering: Python; Go; TypeScript; SQL; FastAPI; Azure; CI/CD; SLURM; 680+ automated tests

Evaluation & statistics: Cross-subject validation; intraclass correlation (ICC); Wilcoxon tests; multiple-comparison correction; nonlinear time-series and dwell-time analysis

EDUCATION

PhD, University College Cork In progress
Neural engineering and machine learning. Supervisor: Luca Longo.

MSc Computer Science, Technological University Dublin Completed 2025
Machine learning, deep learning, data mining, visualisation and statistics. 4.0/4.0 GPA (80/100).

BSc Information Technology, University of Eswatini
Second Class Honours, Upper Division (2.1).

EXPERIENCE

PhD researcher In progress
Artificial Intelligence and Cognitive Load Research Lab (AICL), University College Cork

- Developed two deep-learning approaches for identifying recurring brain-activity patterns, one using image-based maps and another using the brain's electrode network. In a synthetic benchmark, the electrode-network approach matched the known microstate maps exactly and explained 85% of the variance in the test recordings.
- Evaluated 4,832 model configurations using 110 measures of model fit, clustering quality, and consistency across participants, running the analysis on SLURM GPU and IBM Power9 clusters.
- In a single-participant case study, the recurrence measure called determinism was 0.83 for recorded brain activity and 0.60 for phase-randomised controls. The brain-state label sequence showed no evidence of memory beyond the previous state.

Data scientist to Founding engineer May 2021 to present
Etihuku (Curnance)

- Shipped seven production services covering payments, account security, identity checks, background processing, a 120-screen mobile app, an admin dashboard, and a public website.
- Fixed a live withdrawal-fraud issue by ensuring accounts were charged before payouts, with safeguards against double spending and duplicate payments.
- Built a fraud and anti-money-laundering system with 38 rule-based checks, AI-drafted suspicious-activity reports, and privacy controls that hide personal details.
- Built an Azure-based AI document service that reduced manual document creation time by 85% across three compliance regions.

PUBLICATIONS

Peer-reviewed

1. Temporal Dynamics and Latent-Space Interpretability of Generative EEG Microstate Segmentation.
   Saheed Faremi, Andrea Visentin, Luca Longo.
   AICS 2026, University College Dublin (accepted; proceedings to appear), 2026.
2. Integrating Convolutional Variational Autoencoders and the Gaussian Mixture Model for efficient manifold learning and clustering of spatially preserved EEG topographic maps.
   Saheed Faremi, Luca Longo.
   Brain Informatics, 2026. DOI: 10.1186/s40708-026-00327-9.
3. Machine Learning Models for Identifying Factors Influencing and Predicting Malaria Among Children Under Five Years in Nigeria.
   Akinpelumi Saheed Faremi, Boluwaji Akinnuwesi, Elliot Mbunge, Petros M. Mashwama, Stephen Fashoto, Polite Zenzo Ncube, John Batani, Shamsudeen Ademola Sanni, Yinusa A. Faremi, Andile Metfula.
   IEEE ICTAS 2024, 2024. DOI: 10.1109/ICTAS59620.2024.10507142.

Preprints, registered protocols & doctoral work

4. Interpretable EEG Microstate Discovery via Variational Deep Embedding: A Systematic Architecture Search with Multi-Quadrant Evaluation.
   Saheed Faremi, Andrea Visentin, Luca Longo.
   XAI 2026, Late-breaking work and Doctoral Consortium track, Fortaleza, Brazil (arXiv preprint), 2026. DOI: 10.48550/arXiv.2605.10947.
5. Autoencoder-Based Models for Scalp EEG: A Systematic Review of Architectures, Applications, Latent Representations, Interpretability, and Validation.
   Morteza Akbari, **Saheed Akinpelumi Faremi**, Luca Longo.
   OSF Registries (registered systematic-review protocol), 2026. DOI: 10.17605/osf.io/v2w3z.
6. Explainable Disentangled Representation Learning of Recurring Brain Activation Patterns via Variational Autoencoders.
   Saheed Faremi.
   XAI World Conference 2025, Doctoral Proposals track, 2025.

Under review and manuscripts

7. Convolutional Variational Deep Embedding Produces Canonical EEG Microstate Maps and Richer Transition Dynamics than Modified K-Means on Resting EEG.
   Saheed Faremi, Andrea Visentin, Luca Longo.
   Cognitive Computation (manuscript in preparation), 2026.
8. Spatial Encoding Geometry Determines the Maps and Transitions Recovered by Deep EEG Microstate Models.
   Saheed Faremi et al.
   ICLR 2027 (under review; OpenReview #50003), 2026.
9. Prediction of Early Detection of Autism Spectrum Disorder in Toddlers Using Explainable Artificial Intelligence.
   Stephen Fashoto et al.
   Under review, 2026.

TALKS

- Temporal Dynamics and Latent-Space Interpretability of Generative EEG Microstate Segmentation.
  34th Irish Conference on Artificial Intelligence and Cognitive Science (AICS 2026), University College Dublin, 2026.
- Interpretable EEG Microstate Discovery via Variational Deep Embedding.
  XAI 2026, Late-breaking work and Doctoral Consortium track, Fortaleza, Brazil, 2026.
- Machine Learning Models for Predicting Malaria in Nigerian Children Under Five.
  IEEE ICTAS 2024, 2024.